

Noise implementations and lesion–noise experiments

Noise implementation	Configuration	Status	Result or work remaining
Trial framework and seed contract	Repeated trials with separate stimulus and noise seeds; reports mean and variability	Completed	Infrastructure is active. Required outputs are motion-letter d' , $SD(d')$, and $SD(\text{center opponent})$.
V1 Site-2 independent Gaussian noise	Fixed $\sigma = 0.05$ added to the V1 normalization numerator; $N = 50$	Completed, Phase A	Healthy $d' 4.388 \pm 0.033$; amplitude-lesion $d' 3.814 \pm 0.083$. The lesion gap grew from 0.087 without noise to 0.574 with noise; lesion SD was 2.5× healthy.
V1 Site-2 spatially correlated Gaussian noise	$\sigma = 0.05$; Gaussian correlation $\sigma_{\text{corr}} = 3 \text{ px}$; $N = 20$	Completed, Phase B	Healthy $d' 2.864 \pm 0.142$; amplitude-lesion $d' 1.008 \pm 0.153$. Mean gap -1.86 ; SD ratio 1.08. Spatial correlation preserved the lesion ranking but substantially increased its mean effect.

V1 Site-2: uniform vs patchy amplitude	Correlated Gaussian noise; matched mean gain 0.5036; N = 20	Completed	Healthy 2.864; uniform 1.020; patchy 1.015 d'. DIVERGE: NO. V1's normalization denominator is spatially an identity.
MT Site-2: uniform vs patchy amplitude	V1 noise off; correlated noise added at MT numerator; $\sigma =$ 0.05; N = 20	Completed	Healthy $4.468 \pm$ 0.026; uniform 4.362 ± 0.030 ; patchy $4.328 \pm$ 0.030. DIVERGE: NO. The V1- calibrated σ had little effect at MT.
Midget knockout + V1 Site-2 noise	Midget gain = 0; Gaussian V1 noise; 1 and 5°/s; N = 20	Completed	At 1°/s, the healthy-midget gap was 0.210 without noise and 0.215 with noise; SD ratio 0.99. NOISE_AMPLIFIE S: NO. At 5°/s, the noisy gap was 0.047.

Uniform amplitude + delay with noise	Combined deterministic lesion plus a noise site	To be completed if needed	The deterministic combined condition is complete and behaves like amplitude alone. No corresponding noisy combined trial is reported. Current priority is Site 1 and Site 3 rather than repeating Site 2.
Site-1 noise	Noise within shClassV1Basis, representing RGC/optic-nerve variability	To be completed next	Use response-proportional variance initially. Must be implemented and calibrated separately from Sites 2 and 3.
Site-3 noise	Late/readout or decision noise	To be completed next	Must be implemented and calibrated separately from Sites 1 and 2.
Spatially mixing normalization denominator	A denominator D D that mixes damaged and intact spatial locations	To be completed only as an explicitly different experiment	Existing V1 and MT denominators are spatial identities, so the predicted uniform-versus-patchy divergence has not been tested at a genuinely spatially mixing denominator.

Trial-to-trial temporal jitter	Random delay redrawn for each trial or fiber	To be completed	Expected to reduce motion-letter d' more than a fixed random-delay map of the same variance.
Stochastic dropout	Multiplicative, all-or-none conduction failures persistent across a burst	To be completed	Deferred until the individual noise sites have been evaluated. Must not be approximated as Gaussian noise.
Combined Site 1 + Site 2 + Site 3 noise	Multiple noise sources operating simultaneously	To be completed later	Each site must first be tested separately so its mechanism and scale remain interpretable.
VEP/latency output	Time-to-peak, onset latency, or cross-correlation lag	To be defined and completed	The model currently has no locked observable for clinical VEP delay, so no latency result in milliseconds should yet be reported.